

Python Syntax Cheat Sheet (1-Page Note)

Python is a beginner-friendly programming language with clear, readable syntax. This cheat sheet covers the most important fundamentals in one page.

1. Basic Structure

```
print("Hello, Python!")
```

- Python uses indentation (spaces) instead of braces {}.
 - Recommended indentation: 4 spaces
-

2. Variables

```
name = "Minu"
```

```
age = 25
```

```
pi = 3.14
```

```
is_active = True
```

- No need to declare type
 - Types change automatically
-

3. Data Types

```
x = 10          # int
```

```
y = 3.5         # float
```

```
name = "Python" # str
```

```
items = [1, 2, 3] # list
```

```
person = {"name": "A"} # dict
```

4. Input

```
name = input("Enter your name: ")
```

5. Conditional Statements

```
if age > 18:
    print("Adult")
elif age == 18:
    print("Just 18")
else:
    print("Minor")
```

6. Loops

For Loop

```
for i in range(5):
    print(i)
```

While Loop

```
count = 1
while count <= 5:
    print(count)
    count += 1
```

7. Functions

```
def greet(name):
    return "Hello " + name
```

```
print(greet("Minu"))
```

8. Lists

```
fruits = ["apple", "banana", "mango"]
fruits.append("orange") # add
fruits[1] = "grape"     # update
```

9. Dictionaries

```
student = {"name": "Minu", "age": 22}
print(student["name"])
student["age"] = 23
```

10. Exception Handling

```
try:
    x = 10 / 0
except ZeroDivisionError:
    print("Cannot divide by zero")
```

11. Classes & Objects

```
class Student:
    def __init__(self, name):
        self.name = name
```

```
s = Student("Minu")
print(s.name)
```

12. File Handling

```
with open("data.txt", "r") as f:
    print(f.read())
```

Summary

- Python uses indentation instead of braces
- Everything is an object
- Lists, dictionaries, functions, and loops are essential building blocks
- Clean, readable syntax makes Python ideal for beginners

